

Sourav Pal

CS PhD Student at University of Wisconsin-Madison

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Education

University of Wisconsin-Madison

PhD in Computer Sciences

Minor in Mathematics

Advisor: Prof. Vikas Singh

Aug 2019 - Nov 2025 (expected)

Indian Institute of Technology (IIT) Kharagpur

B.Tech. (Hons.) in Computer Science and Engineering

CGPA: 9.33/10.0

Jul 2014 - May 2018

Research Interest

My research is motivated by practical applications and lies at the intersection of deep neural networks and applied mathematics specifically numerical linear algebra and differential equations.

Papers

- **Sourav Pal**, Kamyar Azizzadenesheli, Vikas Singh; “Solving PDE via learnable quadrature” (Under review)
- **Sourav Pal**, Harshavardhan Adep, Clinton Wang, Polina Golland, Vikas Singh; “Implicit Representations via Operator Learning” (ICML 2024, [Paper Link](#))
- Jurijs Nazarovs, Zhichun Huang, Xingjian Zhen, **Sourav Pal**, Rudrasis Chakraborty, Vikas Singh, “Variational Sampling of Temporal Trajectories” (arXiv 24, [Paper Link](#))
- **Sourav Pal**, Zhanpeng Zeng, Sathya N. Ravi, Vikas Singh; “Controlled Differential Equations on Long Sequences via Non-standard Wavelets” (ICML 2023, [Paper Link](#))
- Zhanpeng Zeng, **Sourav Pal**, Jeffery Kline, Glenn Fung, Vikas Singh; “Multi Resolution Analysis (MRA) for Approximate Self-Attention” (ICML 2022, [Paper Link](#))
- Ronak Mehta*, **Sourav Pal***, Vikas Singh, Sathya N. Ravi; “Deep Unlearning via Randomized Conditionally Independent Hessians” (CVPR 2022, [Paper Link](#))
- Gupta Prakhar, **Sourav Pal***, Shubh Gupta*, Ajaykrishnan Jayagopal*, and Ritwik Sinha. “Saliency Prediction for Mobile User Interfaces.” (WACV 2018, [Paper Link](#))
- **Sourav Pal***, Tharun Mohandoss*, Pabitra Mitra. “Visual Attention for Behavioral Cloning in Autonomous Driving.” (ICMV 2018, [Paper Link](#))
- Ankan Mullick, **Sourav Pal**, Projjal Chanda, Arijit Panigrahy, Anurag Bharadwaj and Siddhant Singh. “D-FJ: Deep Neural Network Based Factuality Judgment” (SIGKDD-W 2019 [Paper Link](#))

* Authors contributed equally.

Patents

- Ritwik Sinha, Sourav Pal, Prakhar Gupta, Shubh Gupta and Ajaykrishnan Jayagopal. “Saliency Prediction for a Mobile User Interface.” [US Patent 10,664,999](#) (Granted)
- Ritwik Sinha, Sourav Pal, Prakhar Gupta, Shubh Gupta and Ajaykrishnan Jayagopal. “Saliency Prediction for Informational Documents.” [US Patent 11,263,470](#) (Granted)

Internship

Microsoft Research, Redmond

Jun’22 - Aug’22

Research Intern

Advisor: Dr. Vibhav Vineet, Dr. Neel Joshi

- Worked on synthetic data generation for training large models on computer vision tasks like object detection and segmentation.
- Automate a simple cut-paste strategy of placing varied objects on varying background involving bilevel optimization of the end task and placement strategy.

Adobe

May’21 - Aug’21

Data Science Intern

- Worked on controlled text summarization task.
- Proposed a novel pipeline for the task leveraging on the recent state of the art techniques for the problem.
- Implemented the end to end pipeline.
- Improved already existing methods to improve performance in the task of controlled text generation.

Adobe Research, BigData Experience Lab

May’17 - Jul’17

Research Intern

Advisor: Dr. Ritwik Sinha

- Worked on analysis and improvement of mobile user interfaces using eye gaze tracking.
- Used computer vision techniques to develop deep learning models to predict visual saliency of mobile interfaces.
- Developed metrics for evaluating saliency on a new kind of data set, which was collected by deploying an application in Amazon Mechanical Turk.

Presented the work as a talk at the Adobe Tech Summit 2019 (Feb'3-7), San Francisco.

National Digital Library of India, MHRD Government of India.

May'16 - Jul'16

Summer Intern

Advisor: Prof. Partha Pratim Das, IIT Kharagpur

- Used dynamic instrumentation for debugging and analysis of multithreaded applications.
- Developed algorithms to create tools for data-race detection, deadlock detection, thread-model detection.
- Integrated the above tools with the popular IDE Eclipse as Plug-ins and with Code-blocks as user-defined tools.

Work Experience

Adobe Inc.

Jul'18 - Jul'19

Member of Technical Staff, Document Cloud - Adobe Acrobat

- Worked on prototyping a new feature for the Adobe Acrobat Reader. This feature enabled providing intelligent insights around conversations using machine learning.

The prototype was declared as the winner of the Adobe Document Cloud HackWeek in Noida.

- Worked on voice first documents for Acrobat Reader and integrated the application with Amazon Alexa.

Technical Skills

Programming Languages:

- **Proficient:** Python, Matlab, C++
- **Intermediate:** HTML, Bash, Verilog(HDL), Julia

Tools & Frameworks: PyTorch, Jax, Keras, Numpy, Scikit-learn, OpenCV, Weka, NLTK, L^AT_EX

Relevant Coursework

Mathematical Foundation of Machine Learning	Methods of Applied Mathematics	Real Analysis
Learning Based Computer Vision	Matrix/Linear Algebra	Complex Analysis
Advanced Machine Learning	Probability & Statistics	Measure Theory
Natural Language Processing	Non-Linear Optimization	
Methods of Computational Mathematics	Partial Differential Equations	

Academic Achievement

- Secured an **All India Rank(AIR) of 411 in JEE Main 2014** out of 1.5 million candidates.
- Secured a rank of **51 in WBJEEM 2014** out of 1,00,000 candidates.

Teaching Experience

- Head Teaching Assistant for Data Programming 1 (CS220) Fall 2020 at UW-Madison
- Teaching Assistant for Data Programming 1 (CS220/301) Spring 2020/Fall 2019 at UW-Madison

Reviewing

ICLR 25, ICML 24, NeurIPS 23, ICML 22, CVPR 22, WACV 22

Service

- Mentored undergraduate students at **UW-Madison** as part of the [WISCERS](#) program
- Mentored a group of six undergraduate students from 2016-2018 at **IIT Kharagpur**.
- Active member of **National Service Scheme(NSS)**. Awarded **Certificate of Recognition** for two years of extraordinary social service.
- **School Captain** of The Assembly of God Church School, Purulia. Successfully conducted all student activities during the academic year.